

Digital Face Makeup by Example

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Abstract

This paper introduces an approach of creating face makeup upon a face image with another image as the style example. Our approach is analogous to physical makeup, as we modify the color and skin detail while preserving the face structure. More precisely, we first decompose the two images into three layers: face structure layer, skin detail layer, and color layer. Thereafter, we transfer information from each layer of one image to corresponding layer of the other image. One major advantage of the proposed method lies in that only one example image is required. This renders face makeup by example very convenient and practical. Equally, this enables some additional interesting applications, such as applying makeup by a portraiture. The experiment results demonstrate the effectiveness of the proposed approach in faithfully transferring makeup.

1. Introduction

Face makeup is a technique to change one's appearance with special cosmetics such as foundation, powder, cream *etc.* In most cases, especially for females, makeup is used to enhance one's appearance. With physical face makeup, the foundation and loose powder are usually used to change the texture of face's skin. Foundation is mainly used to conceal flaws and cover the original skin texture, while the loose powder is for introducing new, usually pleasant, texture to skin. Afterwards, applications of other color makeup, such as rouge, eye liner and shadow, *etc.*, follow on top of the powder.

Consider this scenario: when a customer enters a beauty salon, she selects an example image from a catalog and tells the makeup artist to apply the same makeup on her. Before actual task, it would be extremely helpful if she can preview the makeup effects on her own face. However, this is difficult. Traditionally, people have two choices for trying out makeup. One is to physically apply the makeup, which

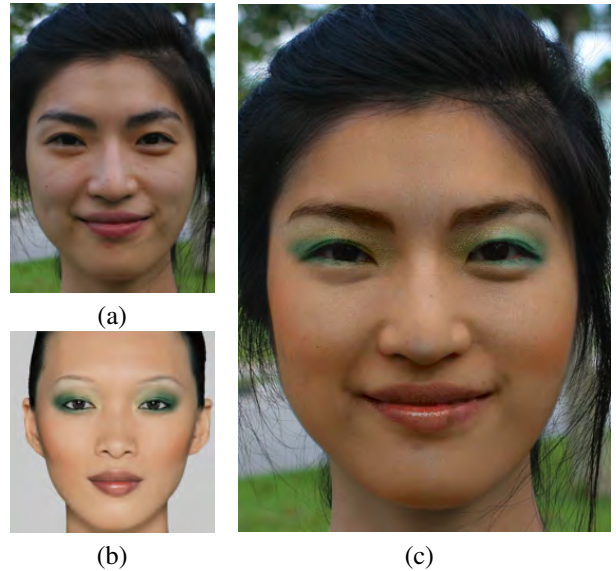


Figure 1. Face makeup by example. (a) A subject image, taken by a common user. (b) An example style image, taken from a professional makeup book [9]. (c) The result of our approach, where foundation effect, eye shadow, and lip highlight in (b) are successfully transferred to (a).

is time-consuming and requires the patience of the participants. Alternatively, one may try on makeup digitally by way of digital photography and with the help of photo editing software, such as Adobe PhotoshopTM. But using such photo editing software is tedious and relies heavily on the users' expertise and effort.

In this paper, we present an approach of creating makeup upon a face image (Fig. 1(a)) with the prototype of another image (Fig. 1(b)) as the style example. This is very practical in the scenario of the beauty salon.

Our approach is inspired by the process of physical makeup. First, we decompose the subject and example images into three layers separately: face structure layer, skin detail layer, and color layer. Ideally, the face structure layer contains only the structure of every face component, such

数字人脸美妆示例法

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摘要

This paper introduces an approach of creating face makeup upon a face image with another image as the style example. Our approach is analogous to physical makeup, as we modify the color and skin detail while preserving the face structure. More precisely, we first decompose the two images into three layers: face structure layer, skin detail layer, and color layer. Thereafter, we transfer information from each layer of one image to corresponding layer of the other image. One major advantage of the proposed method lies in that only one example image is required. This renders face makeup by example very convenient and practical. Equally, this enables some additional interesting applications, such as applying makeup by a portraiture. The experiment results demonstrate the effectiveness of the proposed approach in faithfully transferring makeup.

1. 引言

面部化妆是一种使用特殊化妆品（如粉底、散粉、面霜）改变外貌的技术。在大多数情况下，尤其是女性，化妆用于提升个人的外观。通过物理性面部化妆，通常会先使用粉底和散粉来改变脸部皮肤的质感。粉底主要用于遮盖瑕疵并覆盖原有的肌肤纹理，而散粉则用来为皮肤引入新的、通常是令人愉悦的质感。之后，在散粉之上还会应用其他色彩化妆品，如腮红、眼线和眼影等。

考虑这种情况：当顾客进入美容院时，她会从目录中选择一张示例图片，并告诉化妆师在她的脸上使用相同的妆容。在实际操作之前，如果能够预览她在自己脸上的妆效将会非常有帮助。然而，这很难实现。传统上，人们有两种方式尝试化妆品。一种是物理地将化妆品涂抹在脸上，

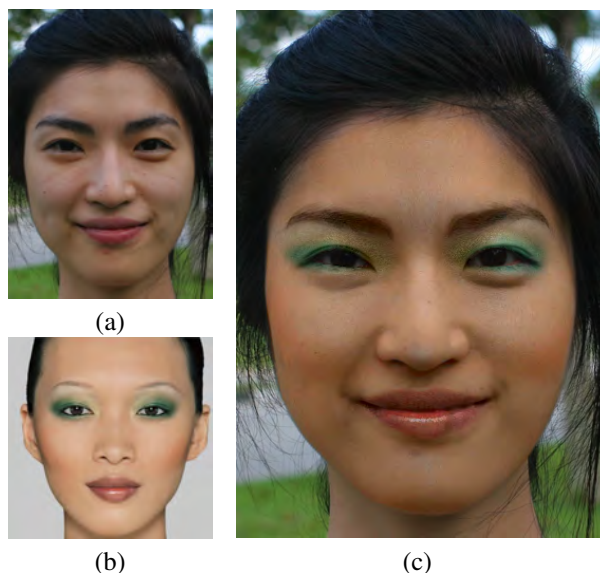


图1. 示例面部妆容。(a) 普通用户拍摄的主体图像。(b) 来自专业化妆书籍[9]的示例风格图像。(c) 我们方法的结果，其中(b)中的底妆效果、眼影和唇部高光成功转移到(a)中。

化妆耗时且需要参与者有足够的耐心。另一种方法是通过数字摄影，并借助如Adobe Photoshop等照片编辑软件来尝试虚拟试妆。但使用此类照片编辑软件既繁琐，又高度依赖用户的技能和努力。

在本文中，我们提出了一种基于另一张图像的原型（图1(b)）作为风格示例，在面部图像（图1(a)）上创建妆容的方法。这种方法在美容院场景中非常实用。

我们的方法借鉴了物理构成的过程。首先，我们将主题图像和示例图像分别分解为三层：面部结构层、皮肤细节层和色彩层。理想情况下，面部结构层仅包含每个面部组件的结构信息。

as the eyes, nose, mouth, etc. The skin detail layer contains the skin texture, including flaws, moles, as well as any wrinkles. The color layer represents color alone. After the three layers are decomposed, face makeup by example can be considered as transferring the skin detail layer and color layer from the makeup example to the subject image while preserving the face structure layer of the subject image.

Contribution. Our proposed approach is effective in transferring face makeup from an example image. This significantly reduces the effort for previewing makeup effects using traditional methods. Moreover, only one single example image is required. This renders face makeup by example much more convenient and practical, compared to previous work which usually requires a pair of “before”-“after” makeup images as examples.

2. Related Work

There is not much previous work addressing digital face makeup. The most closely related work is that of Tong *et al.* [15]. In their work, the way makeup changes the appearance is learned from a pair of example images of the same face “before” and “after” makeup. The quotient of the “after” divided by the “before” is used to represent the change. Then the quotient is multiplied by another image in order to achieve the makeup result. In contrast, our approach requires only one “after” example. This is more convenient and practical, as providing the “before” image is rather difficult in most cases. In addition, it is quite common in the makeup to change the texture of face skin (conceal the original and introduce new one). Because the original texture varies from person to person, the change from “before” to “after” is different between different faces. Thus, it is inappropriate to apply the change across two faces. In contrast, our approach directly transfer the skin texture of the example to the subject image, concealing the original texture. Note that our approach can also keep the texture of the subject image if needed.

Another approach by Ojima *et al.* [10] also uses a pair of “before”-“after” makeup examples. However, only the foundation effect is addressed in their work. In contrast, Tsumura *et al.* [16] employed a physical model to extract hemoglobin and melanin components. Changes in facial appearance are simulated by adjusting the amount of hemoglobin and melanin. The effects they demonstrated include tanning, reddening due to alcohol consumption, aging and cosmetic effects. However, the cosmetic effects are quite limited, and are much simpler than real makeup. Besides, an online commercial software, Taaz [13], provides users with virtual makeup on face photos by simulating the effects of specified cosmetics.

Besides makeup, some existing works also focus on the beautification of face photos. For example, Brand and Pletscher [2] proposed an automatic face photo retouching

method aiming to detect and remove flaws, moles, and acne from faces. Another interesting work by Leyvand *et al.* [6] introduced a technique of modifying face structure to enhance the attractiveness. However, this may also change the identity of the subject, as face structure is usually considered to be a key representation of identity. Conversely, we achieve the goal of beautification by modifying only skin detail and the color, while faithfully preserving the face structure.

The idea of image processing by example can be found in image analogies [5]. Image analogies provide a general framework of rendering an image in different styles. This method learns how pixel values change from a pair of “before”-“after” images as example. This idea was used in Tong *et al.*’s work [15]. As mentioned earlier, the difference is that our approach learns the effects after alteration, while their approach learns the way of altering image pixels.

Our method can also be considered as texture and color transfer. Some previous work on texture transfer is also related. Shan *et al.* [12] proposed a method to transfer the texture across two images. Their method of transferring fine texture is similar to ours in spirit. They used the quotient of the original image and a Gaussian smoothed version of the image to represent the texture. The quotient is multiplied to the smoothed version of another image. This is similar to our layer decomposition. However, there are many differences. First, the Gaussian blur they used may produce halo effect at the strong edges. In contrast, we use an edge-preserving smooth [4] to separate the layers, which successfully suppresses the halo effects. Moreover, they only focus on texture transfer; but it is also important to keep color consistent. We separately transfer the skin detail in lightness channel and color in color channel. In addition, they focused on transferring texture, while our goal is to transfer makeup effect, which is more complex.

3. Digital Makeup

The input of our approach are, a subject image $\mathcal{I}$, a face image to be applied with makeup, and an example image $\mathcal{E}$, providing makeup example. The output is result image $\mathcal{R}$, in which the face structure of $\mathcal{I}$ is retained while the makeup style from $\mathcal{E}$ is applied.

The workflow of our approach is illustrated in Fig. 2. There are four main steps. First, face alignment should be done between the subject and example images. Because the information is transferred pixel by pixel, a fully alignment is necessary before transferring. Followed is layer decomposition (Section 3.2). Both $\mathcal{I}$ and $\mathcal{E}$ are decomposed into three layers: face structure layer, skin detail layer, and the color layer. Third, information from each layer of $\mathcal{E}$ is transferred to corresponding layer of $\mathcal{I}$ in different fashions: skin detail is transferred in an additive way (Section 3.3); color is transferred by alpha blending (Section 3.4); highlight and

皮肤细节层包含皮肤纹理，包括瑕疵、痣以及皱纹等。色彩层仅表示颜色信息。在分解出这三个层次之后，通过示例进行面部化妆可以视为将示例中的皮肤细节层和色彩层转移到目标图像中，同时保留目标图像的面部结构层。

贡献。我们提出的方法在将妆容从示例图像转移到目标面部方面非常有效。这显著减少了使用传统方法预览化妆效果所需的努力。此外，只需要一张单一的示例图像即可完成任务。与以往通常需要一对“妆前”和“妆后”示例图像的工作相比，这种方法使通过示例行化妆变得更加方便和实用。

2. 相关工作

此前关于数字面部化妆的研究较少。与之最为相关的工作是 Tong 等人 [15] 的研究。在他们的工作中，通过一对同一张脸“妆前”和“妆后”的示例图像来学习化妆品如何改变外观。“妆后”图像除以“妆前”图像的商用来表示这种变化。然后将该商乘以另一幅图像以实现化妆效果。相比之下，我们的方法只需要一个“妆后”示例图。这在实际应用中更为方便和实用，因为在大多数情况下提供“妆前”图像较为困难。此外，在化妆过程中经常需要改变面部皮肤的纹理（遮盖原有纹理并引入新的纹理）。由于每个人的原始纹理不同，“妆前”到“妆后”的变化对于不同的脸来说也各不相同。因此，将这种变化应用于两张不同的脸上是不合适的。相比之下，我们的方法直接将示例图像中的皮肤纹理转移到目标图像上，从而遮盖原有的纹理。需要注意的是，如果需要，我们的方法也可以保留目标图像的原有纹理。

另一种方法由 Ojima 等人提出 [10]，也使用了一对“前”与“后”的妆容示例。然而，他们的工作仅关注了基础效果。相比之下，Tsumura 等人 [16] 采用了一个物理模型来提取血红蛋白和黑色素成分。通过调整血红蛋白和黑色素的量，可以模拟面部外观的变化。他们展示的效果包括晒黑、饮酒后的泛红、衰老以及化妆品效果。然而，这些化妆品效果相当有限，并且远没有真实的化妆复杂。此外，一个在线商业软件 Taaz [13] 为用户提供了一种在面部照片上虚拟化妆的功能，通过模拟指定化妆品的效果实现这一目的。

除了化妆之外，一些现有的研究还关注面部照片的美化。例如，Brand 和 Pletscher [2] 提出了一种自动面部照片润饰方法。

一种旨在检测和去除面部瑕疵、痣和痤疮的方法。Leyvand *et al* [6] 提出了一种修改面部结构以增强吸引力的技术。然而，这可能会改变主体的身份，因为面部结构通常被认为是身份的关键表现形式。相反，我们通过仅修改皮肤细节和颜色来实现美化目标，同时忠实保留面部结构。

基于示例的图像处理思想可以在文献 [5] 中的图像类比方法中找到。图像类比提供了一种以不同风格渲染图像的一般框架。该方法通过一对“之前”和“之后”的图像作为示例，学习像素值的变化规律。这一思路被 Tong *et al* 在文献 [15] 的工作中采用。如前所述，区别在于我们的方法学习的是改变后的效果，而他们的方法则是学习如何改变图像的像素值。

我们的方法也可以被视为纹理和颜色的转移。一些先前关于纹理转移的工作与此相关。Shan *et al* [12] 提出了一种在两张图像之间传递纹理的方法。他们用于传递细纹理的方法与我们有相似之处。他们使用原始图像与其高斯平滑版本相除来表示纹理，然后将该商数乘以另一张图像的平滑版本。这类似于我们的层分解方法。然而，两者存在许多不同之处。首先，他们使用的高斯模糊可能会在强边缘处产生晕轮效应，而我们采用了一种保持边缘的平滑算法 [4] 来分离层次，从而成功抑制了晕轮效应。此外，他们的工作仅关注纹理转移；但保持颜色一致性也很重要。我们在亮度通道中单独传递皮肤细节，在色彩通道中传递颜色。另外，他们专注于纹理转移，而我们的目标是传递妆效，这更为复杂。

3. 数字美颜技术

我们的方法输入包括一张目标图像 I 、一张待添加妆容的面部图像以及提供妆容示例的参考图像 E 。输出为结果图像 R ，其中保留了 I 的面部结构，并应用了来自 E 的妆容风格。

我们的方法的工作流程如图 2 所示。主要分为四个步骤。首先，需要在主体图像和示例图像之间进行面部对齐。由于信息是逐像素传输的，在传输之前必须完成完全对齐。接下来是层分解（见 3.2 节）。 I 和 E 均被分解为三层：面部结构层、皮肤细节层以及颜色层。第三步，来自 E 各个层次的信息以不同的方式传输到 I 的对应层次中：皮肤细节采用累加方式进行传输（见 3.3 节）；颜色通过 alpha 混合进行传输（见 3.4 节）；高光和

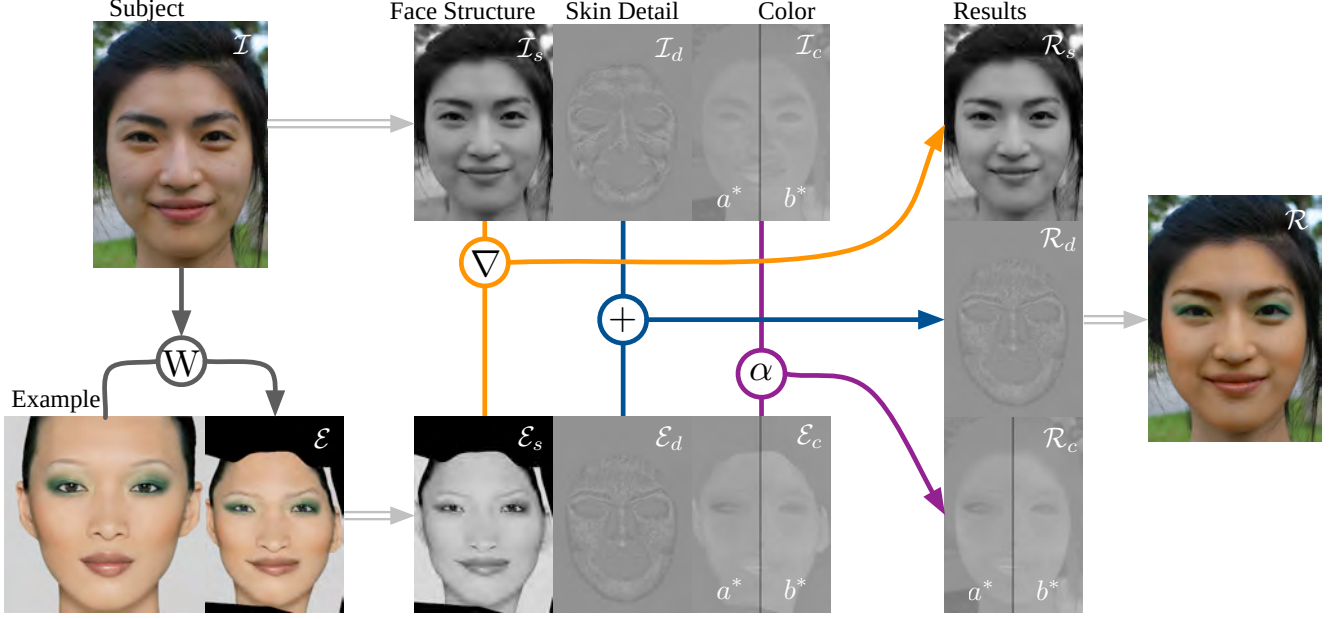


Figure 2. The workflow of our approach. W denotes warping. ∇ denotes gradient editing. $+$ denotes weighted addition. α denotes alpha blending. In this figure, the value of skin detail layer, i.e. $\mathcal{I}_d$ and $\mathcal{E}_d$, is exaggerated 4 times for better visualization. Our approach consists of four main steps: face alignment, layer decomposition, makeup transferring, and layer composition.

shading effects in face structure layer are transferred in the way of gradient editing (Section 3.5). Finally, three resultant layers are composed together.

3.1. Face alignment

For face alignment, we adopt the Thin Plate Spline (TPS) [1] to warp the example image $\mathcal{E}$ to subject image $\mathcal{I}$. The control points required by TPS are obtained using an extended Active Shape Model (ASM) [8]. Due to the diversity of face appearance under various possible makeup, ASM may not get the accurate position of points. Our system may still require user to refine the position of control points. Since the control points have been roughly located already, the refinement does not require much effort. It usually takes less than 1 minute to refine the control points for a face. An example of refined control points is shown in Fig. 3(a). There are 83 points in total on a face.

As shown in Fig. 3(b), these control points have defined different face components, viz eyebrows, eyes, nose, nostrils, lips, mouth cavity (the space between lips), and other facial skin (the rest of the face). These components are further divided into three classes to be treated in different ways during makeup. These three classes ($\mathcal{C}_1 \sim \mathcal{C}_3$) are illustrated in different colors in Fig. 3(b). $\mathcal{C}_1$ (the skin region, the entire face excluding $\mathcal{C}_2$ and $\mathcal{C}_3$) follows the workflow illustrated in Fig. 2. Since the texture of $\mathcal{C}_2$ (lips) varies greatly from person to person and the region of $\mathcal{C}_2$ is easy to deform, we use a special method to transfer the makeup style in this region (discussed in Section 3.6). $\mathcal{C}_3$ (eyes and

mouth cavity) is kept untouched all the time during entire makeup process.

3.2. Layer decomposition

The subject image $\mathcal{I}$ and example image $\mathcal{E}$ (after warping) are first decomposed into color and lightness layers. We then further decompose the lightness layer into face structure and skin detail layers.

In the first step, $\mathcal{I}$ and $\mathcal{E}$ are decomposed into color and lightness layers by converting them to CIELAB colorspace. The L^* channel is considered as lightness layer and a^* , b^* channel the color layer. We choose the CIELAB colorspace because it performs better than other color spaces in separating lightness from color [17], and it is approximately perceptually uniform [7].

Second, the lightness layer is decomposed to large-scale and detail layers. The large-scale layer is considered as the face structure layer and the detail layer as skin detail layer. Large-scale/detail layer decomposition has been addressed in many works, such as [3] and [19]. The main idea is first to perform an edge-preserving smoothing on the lightness layer to obtain the large scale layer, and then to subtract (or divide) the large scale layer from the lightness layer to obtain the detail layer. In this approach, we adapt a weighted-least-squares (WLS) operator recently proposed by Farbman *et al.* [4]. An alternative method is bilateral filtering [14], which was used in many previous works. We choose this WLS operator because of its better performance compared to the bilateral filter, especially when the blur level

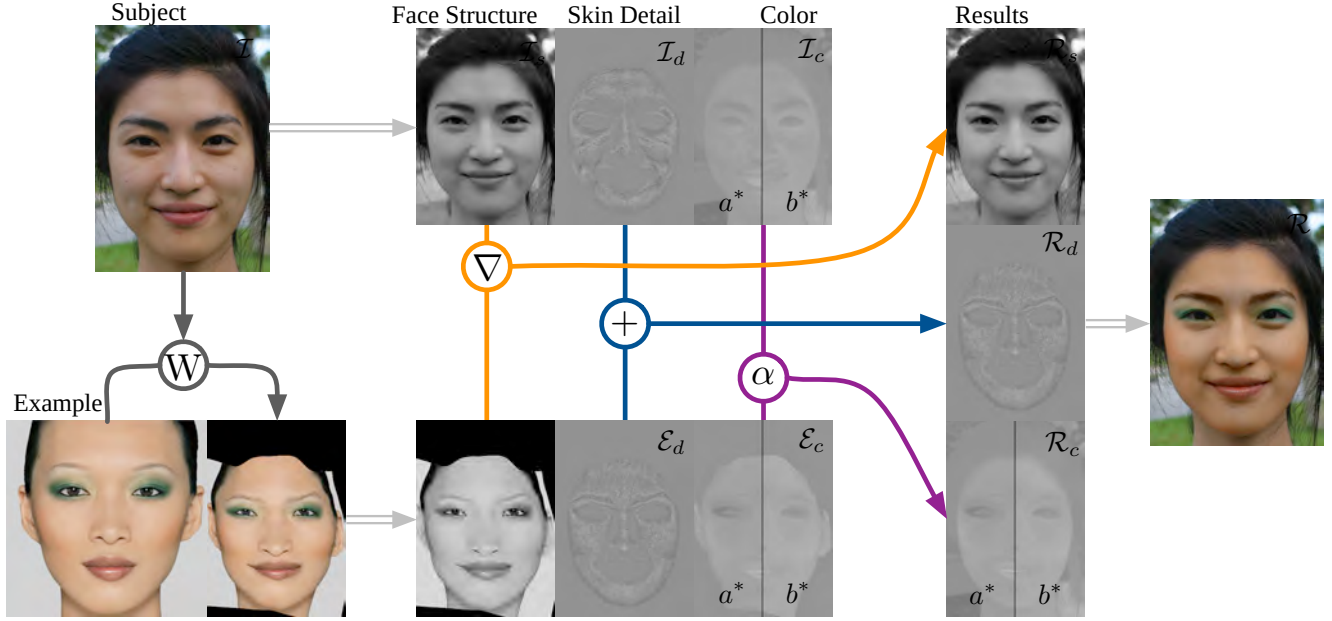


图2 我们方法的工作流程。 W 表示扭曲操作。 ∇ 表示梯度编辑。 $+$ 表示加权求和。 α 表示 alpha 混合。在该图中，皮肤细节层的值 $i.e.$ $\mathcal{I}_d$ 和 $\mathcal{E}_d$ 的数值被放大了四倍以更好地进行可视化展示。我们的方法主要包括四个主要步骤：面部对齐、层次分解、妆容转移和层次合成。

面部结构层中的阴影效果通过渐变编辑方式进行传递（第3.5节）。最后，三个生成的图层被组合在一起。

3.1. 人脸对齐

对于人脸对齐，我们采用薄板样条（TPS）[1]将示例图像 $\mathcal{E}$ 变换到目标图像 $\mathcal{I}$ 。TPS所需的控制点通过扩展的主动形状模型（ASM）[8]获得。由于在各种可能的妆容下人脸外观的多样性，ASM可能无法准确获取点的位置。我们的系统仍需用户微调这些控制点的位置。由于控制点已经大致定位，因此微调工作并不需要太多努力。通常调整一个人脸上的控制点只需不到1分钟的时间。图3(a)展示了经过微调后的控制点示例。人脸总共有83个控制点。

如图3(b)所示，这些控制点定义了不同的面部组件，即眉毛、眼睛、鼻子、鼻孔、嘴唇、口腔（唇部之间的空间）和其他面部皮肤（脸部的其余部分）。这些组件进一步被划分为三类，在化妆过程中以不同方式进行处理。这三类 ($C_1 \sim C_3$) 在图3(b)中用不同的颜色表示。 C_1 (皮肤区域，即整个面部排除掉 C_2 和 C_3) 的部分，遵循图2所示的工作流程。由于 C_2 (唇部) 的纹理因人而异且该区域容易变形，我们使用一种特殊方法在此区域内转移化妆风格（详见第3.6节）。 C_3 (眼睛和

在整个化妆过程中，口腔内部始终保持不变。

3.2. 层分解

待处理的图像 $\mathcal{I}$ 和经过变形后的示例图像 $\mathcal{E}$ (首先被分解为颜色层和亮度层。然后，进一步将亮度层分解为人脸结构层和皮肤细节层。

在第一步中，将 $\mathcal{I}$ 和 $\mathcal{E}$ 转换为 CIELAB 颜色空间，从而将其分解为色彩层和亮度层。其中， L^* 通道被视为亮度层，而 b^* 和 a^* 通道则视为色彩层。我们选择 CIELAB 颜色空间是因为它在分离亮度与色彩方面优于其他颜色空间[17]，并且其感知均匀性近似于理想状态[7]。

其次，亮度层被分解为大尺度层和细节层。大尺度层被视为面部结构层，而细节层则视为皮肤细节层。这种大尺度/细节层的分解在许多文献中都有讨论，例如[3]和[19]。主要思路是首先对亮度层进行边缘保持平滑处理以获得大尺度层，然后从亮度层中减去（或除以）该大尺度层以得到细节层。在此方法中，我们采用 Farbman 等人最近提出的一种加权最小二乘法 (WLS) 算子 [v25][4]。另一种方法是双边滤波[14]，这种方法在许多先前的工作中都有应用。我们选择 WLS 算子是因为它相比双边滤波器具有更好的性能，尤其是在模糊程度较高的情况下。

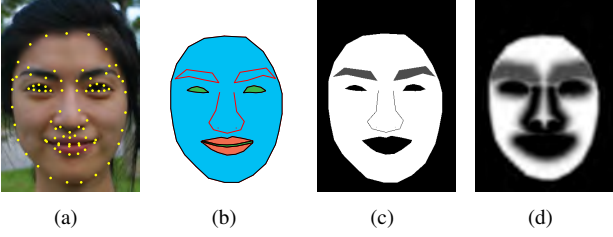


Figure 3. (a) Control points on a face. (b) Facial components defined by control points in (a), including eyebrows, eyes, nose, nostrils, lips, mouth cavity, and other facial skin, which are further divided into three classes, blue as $\mathcal{C}_1$, red as $\mathcal{C}_2$, and green as $\mathcal{C}_3$. (c) Unsmoothed version of β value. (d) Illustration of β value defined by Eq. (3), brighter pixel denoting higher β value.

increases.

Suppose that the lightness layer and the face structure texture layer are denoted by l and s , respectively. The problem of solving s can be formulated as minimization of the energy function:

$$E = |s - l|^2 + \lambda H(\nabla s, \nabla l). \quad (1)$$

The first term $|s - l|^2$ is to keep s similar to l , while the regularization term $H(\nabla s, \nabla l)$ is trying to make s as smooth as possible.

The WLS operator described in [4] performs the same level of smoothing all over the image; but we expect different levels of smoothness in different regions. Thus, a spatial-variant coefficient β is added to H . Then H is defined as

$$H(\nabla s, \nabla l) = \sum_p \beta(p) \left(\frac{|s_x(p)|^2}{|l_x(p)|^\alpha + \epsilon} + \frac{|s_y(p)|^2}{|l_y(p)|^\alpha + \epsilon} \right), \quad (2)$$

where p indexes the image pixel, ϵ is a small constant preventing division by zero, $\{\cdot\}_x$ and $\{\cdot\}_y$ denote the partial derivative of $\{\cdot\}$ along x and y coordinates respectively, while α is a coefficient for adjusting the effect of ∇l on ∇s . We use $\alpha = 1.2$ and $\lambda = 0.2$ in all our experiments.

We expect that $\beta(p)$ is low inside the facial components, and equal to 1 over the skin area. As shown in Fig. 3(c), $\beta(p)$ is 0.3 in the eyebrow region, 0 in other facial component region, and 1 in facial skin. In addition, we also expect $\beta(p)$ changes smoothly over the whole image. Thus, we further define $\beta(p)$ as

$$\beta(p) = \min_q \left(1 - k(q) \cdot e^{-\frac{(q-p)^2}{2\sigma^2}} \right), \quad (3)$$

where q indexes the pixel over the image. $k(q)$ is 0.7 for eyebrows, 0 for skin area, 1 for other facial components. The value of σ^2 is set to $\min(\text{height}, \text{width})/25$. Then, we have $\beta(p)$ as shown in Fig. 3(d).

Since the L^* channel is approximately perceptually uniform, we use subtraction to obtain the skin detail layer d

from lightness layer l , *i.e.*

$$d(p) = l(p) - s(p). \quad (4)$$

Two examples of our decomposition results are shown in Fig. 2. We can see that detail is controlled well by $\beta(p)$. Because β is zero in the eyes and mouth regions and outside the facial region, the skin detail layer there is zero.

In the rest of the paper, we will use $\{\cdot\}_s$, $\{\cdot\}_d$, $\{\cdot\}_c$ to denote $\{\cdot\}$'s the face structure layer, skin detail layer, and color layer, respectively.

3.3. Skin detail transfer

Skin detail transfer is straightforward. The resultant skin detail layer $\mathcal{R}_d$ is a weighted sum of $\mathcal{I}_d$ and $\mathcal{E}_d$, *i.e.*

$$\mathcal{R}_d = \delta_{\mathcal{I}} \mathcal{I}_d + \delta_{\mathcal{E}} \mathcal{E}_d, \quad (5)$$

where $0 \leq \delta_{\mathcal{I}}, \delta_{\mathcal{E}} \leq 1$. The values of $\delta_{\mathcal{I}}$ and $\delta_{\mathcal{E}}$ control the contribution of each component.

For different applications, different $\delta_{\mathcal{I}}$ and $\delta_{\mathcal{E}}$ values can be used. As we mentioned in Section 1, the purpose of foundation and loose powder in physical makeup is to conceal the original skin detail and to introduce new skin detail. Thus, we set $\delta_{\mathcal{I}} = 0$ to conceal $\mathcal{I}_d$, and $\delta_{\mathcal{E}} = 1$ to transfer $\mathcal{E}_d$ to $\mathcal{R}_d$. This is a typical setting for beauty makeup transfer. It is used in all our experiment results except the ones showing different manipulation of makeup effects (Fig. 4). In some cases, we can also set $\delta_{\mathcal{I}} > 0$ to keep some original skin detail. Note that the sum of two weights is not required to be 1 because $\mathcal{R}_d$ can come from any amount of $\mathcal{I}_d$ or $\mathcal{E}_d$. In addition, the sum should not be very small, otherwise the face in result image $\mathcal{R}$ would be not realistic due to lack of skin detail.

3.4. Color transfer

The resultant color layer $\mathcal{R}_c$ is an alpha-blending of color layers of $\mathcal{I}$ and $\mathcal{E}$, *i.e.*

$$\mathcal{R}_c(p) = \begin{cases} (1 - \gamma) \mathcal{I}_c(p) + \gamma \mathcal{E}_c(p) & p \in \mathcal{C}_3 \\ \mathcal{I}_c(p) & \text{otherwise} \end{cases} \quad (6)$$

The value of γ is to control blending effect of two color layers. The result in Fig. 1 is obtained with $\gamma = 0.8$.

3.5. Highlight and shading transfer

The highlight and shading effects of makeup lie in the L^* channel. Because the face structure layer is actually the large scale layer of L^* channel, the smooth change of the highlight and shading effects remain in the face structure layer. Since these effects are important for makeup, we should transfer them across this layer.

Because face structure layer contains identity information, we can neither directly copy $\mathcal{E}_s$ over $\mathcal{I}_s$ nor blend

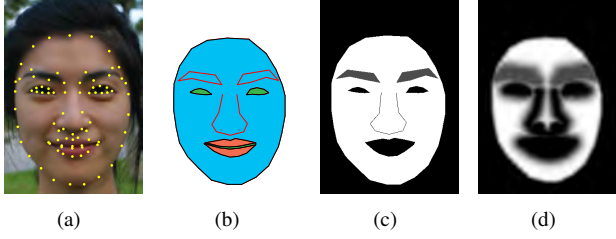


图3. (a) 面部上的控制点。 (b) 由 (a) 中的控制点定义的面部组件, 包括眉毛、眼睛、鼻子、鼻孔、嘴唇、口腔以及其它面部皮肤, 这些组件进一步划分为三类: 蓝色表示 C_1 , 红色表示 C_2 , 绿色表示 C_3 。 (c) 未经平滑处理的 β 值。 (d) 由公式 (3) 定义的 β 值示意图, 亮度更高的像素表示更大的 β 值。

增加。

假设光照层和面部结构纹理层分别用 l 和 s 表示。求解 s 的问题可以表述为能量函数的最小化:

$$E = |s - l|^2 + \lambda H(\nabla s, \nabla l). \quad (1)$$

第一个项 $|s - l|^2$ 是为了保持 s 与 l 相似, 而正则化项 $H(\nabla s, \nabla l)$ 则是为了使 s 尽可能平滑。

《4》中描述的WLS算子在整个图像上执行相同程度的平滑处理; 但我们期望在不同区域具有不同的平滑度。因此, 添加了一个空间变系数 β 到 H 。然后将 H 定义为

$$H(\nabla s, \nabla l) = \sum_p \beta(p) \left(\frac{|s_x(p)|^2}{|l_x(p)|^\alpha + \epsilon} + \frac{|s_y(p)|^2}{|l_y(p)|^\alpha + \epsilon} \right), \quad (2)$$

其中, p 表示图像像素, ϵ 是一个小常数, 用于防止除以零的情况发生, $\{.\}_x$ 和 $\{.\}_y$ 分别表示沿 x 坐标和 y 坐标的 $\{.\}$ 的偏导数, 而 α 是一个调整因子, 用于调节 ∇l 对 ∇s 影响的程度。在我们所有的实验中都使用了 $\alpha = 1.2$ 和 $\lambda = 0.2$ 。

我们预计在面部组件内部 $\beta(p)$ 值较低, 而在皮肤区域则为1。如图3(c)所示, 在眉毛区域 $\beta(p)$ 的值为0.03, 其他面部组件区域为0, 而面部皮肤区域为1。此外, 我们还期望整个图像中 $\beta(p)$ 的变化平滑。因此, 我们进一步定义 $\beta(p)$ 为

$$\beta(p) = \min \left(1 - k(q) \cdot e^{-\frac{(q-p)^2}{2\sigma^2}} \right), \quad (3)$$

其中 q 表示图像中的像素位置。 $k(q)$ 的值为眉毛区域 0.7, 皮肤区域 0, 其他面部组件区域 1。 σ^2 的值被设置为 $\min(\text{高度}, \text{宽度})$ 的一半, 即 $/25$ 。 然后我们有如图 3 (d) 所示的 $\beta(p)$ 。

由于通道 L^* 大致上是感知均匀的, 我们使用减法来获取皮肤细节层 d

从轻量层 l , *i.e.*

$$d(p) = l(p) - s(p). \quad (4)$$

如图2所示, 我们展示了两个分解结果的例子。可以看到细节由 $\beta(p)$ 很好地控制着。由于在眼睛和嘴巴区域以及面部区域之外, β 为零, 因此这些区域的皮肤细节层也为零。

在本文的其余部分, 我们将分别使用 $\{.\}_s$ 、 $\{.\}_d$ 和 $\{.\}_c$ 来表示 $\{.\}$ 的面结构层、皮肤细节层和颜色层。

3.3. 皮肤细节传输

皮肤细节转移很简单。所得的皮肤细节层 $\mathcal{R}_d$ 是 $\mathcal{I}_d$ 和 $\mathcal{E}_d$ 的加权和, 即 *i.e.*

$$\mathcal{R}_d = \delta_{\mathcal{I}} \mathcal{I}_d + \delta_{\mathcal{E}} \mathcal{E}_d, \quad (5)$$

其中 $0 \leq \delta_{\mathcal{I}}, \delta_{\mathcal{E}} \leq 1$ 。 $\delta_{\mathcal{I}}$ 和 $\delta_{\mathcal{E}}$ 的值控制每个组件的贡献程度。

对于不同的应用, 可以使用不同的 $\delta_{\mathcal{I}}$ 和 $\delta_{\mathcal{E}}$ 值。正如我们在第1节中提到的, 粉底和散粉在物理妆容中的作用是遮盖原始皮肤细节并引入新的皮肤细节。因此, 我们将 $\delta_{\mathcal{I}}$ 设置为0以遮盖 $\mathcal{I}_d$, 并将 $\delta_{\mathcal{E}}$ 设置为1以将 $\mathcal{E}_d$ 转移到 $\mathcal{R}_d$ 。这是美妆转移的典型设置, 在我们所有的实验结果中都使用了这一设置 (图4除外, 该图展示了不同妆效操作的结果)。在某些情况下, 也可以将 $\delta_{\mathcal{I}} > 0$ 设置为0以保留一些原始皮肤细节。需要注意的是, 两个权重之和不需要等于1, 因为 $\mathcal{R}_d$ 可以来自任意数量的 $\mathcal{I}_d$ 或 $\mathcal{E}_d$ 。此外, 总和不应太小, 否则结果图像中的脸部由于缺乏皮肤细节而显得不够真实。

3.4. 颜色传递

结果颜色层 $\mathcal{R}_c$ 是对颜色层 $\mathcal{I}$ 和 $\mathcal{E}$ 进行 alpha 混合后的产物, *i.e.*

$$\mathcal{R}_c(p) = \begin{cases} (1 - \gamma) \mathcal{I}_c(p) + \gamma \mathcal{E}_c(p) & p \in C_3 \\ \mathcal{I}_c(p) & \text{otherwise} \end{cases} \quad (6)$$

γ 的值用于控制两个颜色层的混合效果。图1中的结果是在 $\gamma = 0.8$ 时获得的。

3.5. 高亮和阴影传输

化妆的高光和阴影效果体现在 L^* 通道中。由于面部结构层实际上是 L^* 通道的大尺度层次, 因此高光和阴影的平滑变化会保留在面部结构层内。鉴于这些效果对于化妆至关重要, 我们应该将它们传递到这一层。

由于面部结构层包含身份信息, 我们既不能直接将 $\mathcal{E}_s$ 复制到 $\mathcal{I}_s$ 上, 也不能进行融合。

them. Instead, we adapt a gradient-based editing method. The idea is to add only large changes of $\mathcal{E}_s$ to $\mathcal{I}_s$. Doing this, we assume that these changes are due to makeup. This assumption holds if the illumination of $\mathcal{E}$ is approximately uniform.

Gradient-based editing can preserve the illumination of $\mathcal{I}$, transfer the highlight and shading effects, and meanwhile yield smooth result. Editing an image in the gradient domain was introduced by Pérez *et al.* [11] and was employed in many later works. The gradient-based method used here is similar. The gradient of R_s is defined as

$$\nabla R_s(p) = \begin{cases} \nabla \mathcal{E}_s(p) & \text{if } \beta(p) \|\nabla \mathcal{E}_s(p)\| > \|\nabla \mathcal{I}_s(p)\| \\ \nabla \mathcal{I}_s(p) & \text{otherwise} \end{cases} \quad (7)$$

Since only the gradient of the face region ($\mathcal{C}_3$) is changed (but not its boundary nor regions outside $\mathcal{C}_3$), the process of solving the resultant face structure layer $\mathcal{R}_s$ from its gradient is equivalent to solving a Poisson equation with Dirichlet boundary condition. We use the Gauss-Seidel method with successive over-relaxation to solve the Poisson equation.

3.6. Lip makeup

The makeup effect of lip region ($\mathcal{C}_2$) is quite different from that of face skin ($\mathcal{C}_1$). In physical makeup, cosmetics on lips (*e.g.* lipstick) usually preserve or highlight the texture of lips, rather than conceal it as in face skin. Thus, lips region makeup needs to be treated in a different way.

The main idea is to fill each pixel of $\mathcal{R}$ with pixel value from $\mathcal{E}$ guided by $\mathcal{I}$. Then the makeup effect is similar to $\mathcal{E}$ and the texture is similar to $\mathcal{I}$. Specifically, for each pixel p in $\mathcal{I}$, we search for a pixel q in $\mathcal{E}$ so that $\mathcal{E}(q)$ and $\mathcal{I}(p)$ are as similar as possible, while q and p are as close as possible.

Suppose that lip region after makeup is denoted by $\mathcal{M}$. For each $p \in \mathcal{C}_2$, we have

$$\mathcal{M}(p) = \mathcal{E}(\tilde{q}), \quad (8)$$

where

$$\tilde{q} = \arg \max_{q \in \mathcal{C}_2} \{G(|q - p|)G(|\mathcal{E}(q) - \mathcal{I}(p)|)\} \quad (9)$$

where $G(\cdot)$ denotes Gaussian function. For $|\mathcal{E}(q) - \mathcal{I}(p)|$, we use the difference of pixel values in only L^* channel after histogram equalization of $\mathcal{E}$ and $\mathcal{I}$ separately.

The makeup result of lips region $\mathcal{M}$ is then merged to result image $\mathcal{R}$. The L^* channel of $\mathcal{M}$ is added to L^* channel of $\mathcal{R}$ with a gradient replacement. The a^* , b^* channel of $\mathcal{M}$ replace corresponding region in $\mathcal{R}$.

4. Experiments and Results

Beauty makeup: Our approach can manipulate makeup effect in various ways. In case of heavy foundation where

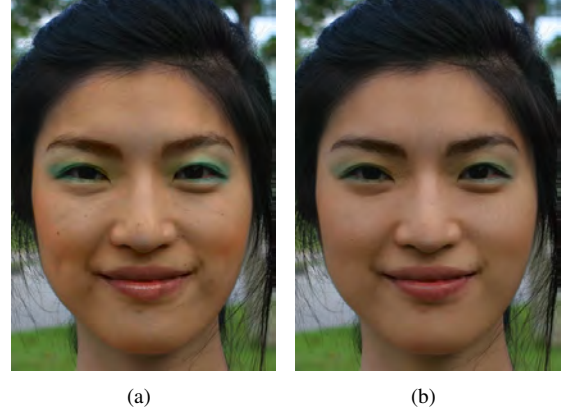


Figure 4. Manipulation of makeup effects. (a) Light foundation effect. (b) Light color effect.

the foundation covers the original skin, we use $\delta_{\mathcal{I}} = 0$, $\delta_{\mathcal{E}} = 1$ to transfer all skin detail from $\mathcal{E}$ to $\mathcal{I}$. An example of result is shown in Fig. 1(c). To simulate light foundation effect, we use $\delta_{\mathcal{I}} = 1$, $\delta_{\mathcal{E}} = 0$ to keep original skin detail. An example is Fig. 4(a). Adjusting $\delta_{\mathcal{I}}$, $\delta_{\mathcal{E}}$ can simulate different level of foundation effect. For a good result of concealing the skin detail of $\mathcal{I}$, it is required that the skin detail of $\mathcal{E}$ should be only due to makeup, *i.e.* $\mathcal{E}$ should have heavy foundation (usually also with loose powder); otherwise $\mathcal{R}$ may contain imperfection of skin detail from $\mathcal{E}$. Another manipulation is on the level of makeup color by adjusting γ . As shown in Fig. 4(b), we used $\gamma = 0.5$ for a light makeup color.

A comparison of our result with that of Tong *et al.* is shown in Fig. 5. Note that they used an additional image as example of “before” makeup, which is not shown here. In contrast, we only use the “after” image. Our result shows vivid color and more similar makeup style to the example image. From the close-up view of the eye, we can see that our approach preserves the structure much better than theirs. Moreover, the highlight of lips in our result is successfully transferred while the texture is preserved. It is much more similar to real makeup. For skin tone, we have different representations. Our result takes the skin tone from example image, while their result keeps the original tone. As we presented before, our aim was to apply the makeup style faithfully from the example image, including color, skin detail. In the physical makeup, tone color is usually also considered to be a part of makeup. Another consideration is that color used in makeup should match the skin tone [18]; makeup transferring without skin tone may destroy the original harmony.

Photo retouching: Our approach can be also used to retouch a face photo. This is almost the same as face makeup by example. We address it separately because this application itself is interesting and useful. Instead of photographs

我们采用基于梯度的编辑方法进行调整。其核心思想是仅对从 $\mathcal{E}_s$ 到 $\mathcal{I}_s$ 的大变化部分进行修改，假设这些变化是由化妆引起的。这一假设成立的前提是 $\mathcal{E}$ 的光照近似均匀。

基于梯度的编辑可以保留光照效果 $\mathcal{I}$ ，传递高光和阴影效果，并且同时产生平滑的结果。在梯度域中编辑图像的概念由Pérez *et al*提出[11]，并在许多后续工作中被采用。这里使用的基于梯度的方法与此类似。 R_s 的梯度定义为

$$\nabla R_s(p) = \begin{cases} \nabla \mathcal{E}_s(p) & \text{if } \beta(p) \|\nabla \mathcal{E}_s(p)\| > \|\nabla \mathcal{I}_s(p)\| \\ \nabla \mathcal{I}_s(p) & \text{otherwise} \end{cases}$$

仅面部区域 (C_3) 的梯度发生变化 (但其边界及外部区域 C_3 不变)，因此从该梯度求解所得的面部结构层 R_s 的过程等同于求解一个具有狄利克雷边界条件的泊松方程。我们采用带有逐次超松弛的高斯-赛德尔方法来求解泊松方程。

3.6. 唇部妆容

唇部区域 (C_2) 的妆效与面部皮肤 (C_1) 有很大不同。在物理化妆中，用于嘴唇的化妆品 (如口红 *e.g*)，通常会保留或突出唇部的纹理，而不是像面部皮肤那样遮盖它。因此，唇部区域的化妆需要采用不同的处理方式。

主要思想是将 $\mathcal{R}$ 中的每个像素填充为来自 $\mathcal{E}$ 的像素值，由 $\mathcal{I}$ 引导。这样，美化效果类似于 $\mathcal{E}$ ，纹理则类似于 $\mathcal{I}$ 。具体而言，对于 $\mathcal{I}$ 中每个像素 p ，我们在 $\mathcal{E}$ 中寻找一个像素 q ，使得 $\mathcal{E}(q)$ 和 $\mathcal{I}(p)$ 尽可能相似，同时 q 和 p 尽可能接近。

假设化妆后的唇部区域用 $\mathcal{M}$ 表示。对于每个 $p \in C_2$ ，我们有

$$\mathcal{M}(p) = \mathcal{E}(\tilde{q}), \quad (8)$$

其中

$$\tilde{q} = \arg \max_{q \in C_2} \{G(|q - p|)G(|\mathcal{E}(q) - \mathcal{I}(p)|)\} \quad (9)$$

其中 $G(\cdot)$ 表示高斯函数。对于 $|\mathcal{E}(q) - \mathcal{I}(p)|$ ，我们在分别对 $\mathcal{E}$ 和 $\mathcal{I}$ 进行直方图均衡化后的 L^* 通道像素值差分中使用。

唇部区域 $\mathcal{M}$ 的上妆结果然后合并到结果图像 $\mathcal{R}$ 中。 $\mathcal{M}$ 的 a^* 通道与 $\mathcal{R}$ 的 L^* 通道通过梯度替换进行叠加。 $\mathcal{M}$ 的 b^* 通道替换对应区域中的 $\mathcal{R}$ 。

4. 实验与结果

美妆修饰：我们的方法可以以多种方式操控妆效。在处理厚重粉底的情况下，



图4. 化妆效果的调整。(a) 素颜提亮效果。(b) 轻薄色彩效果。

基础覆盖了原皮肤，我们使用 $\delta_{\mathcal{I}} = 0, \delta_{\mathcal{E}} = 1$ 将所有皮肤细节从 $\mathcal{E}$ 转移到 $\mathcal{I}$ 。结果示例如图1(c)所示。为了模拟底妆效果，我们使用 $\delta_{\mathcal{I}} = 1, \delta_{\mathcal{E}} = 0$ 来保持原皮肤细节。一个例子如图4(a)所示。调整 $\delta_{\mathcal{I}}, \delta_{\mathcal{E}}$ 可以模拟不同级别的底妆效果。为了获得良好的遮瑕效果，要求 $\mathcal{I}$ 的皮肤细节仅因化妆而产生，即 $\mathcal{E}$ 。 $\mathcal{E}$ 应该有厚重的底妆 (通常还会使用散粉)；否则 $\mathcal{R}$ 可能会包含来自 $\mathcal{E}$ 的皮肤瑕疵。另一种操作是通过调整 γ 来改变化妆颜色的程度，如图4(b)所示，我们使用了 $\gamma = 0.5$ 实现淡妆效果。

我们的结果与 Tong *et al* 的对比如图 5 所示。请注意，他们使用了一张额外的“妆前”图片作为示例，此处未展示。相比之下，我们仅使用了“妆后”图片。我们的结果显示色彩鲜艳且更接近示例图像中的化妆风格。从眼部特写可以看出，我们的方法比他们的方法更好地保留了结构。此外，我们的结果成功地将唇部的高光效果进行了转移，并保持了原有的纹理，更加贴近真实的化妆效果。对于肤色方面，我们有不同的表现方式。我们的结果显示的是从示例图片中提取的肤色，而他们的结果显示则保留了原始肤色。正如之前所述，我们的目标是忠实于示例图像中的妆容风格进行应用，包括颜色和皮肤细节。在实际化妆过程中，色调通常也被视为妆容的一部分[18]；使用与肤色不匹配的颜色可能会破坏原有的和谐感。

照片修图：我们的方法也可以用于修复一张人脸照片。这几乎与示例妆容相同。我们单独处理这一应用，因为该应用本身既有趣又实用。而不是使用照片



Example image $\mathcal{E}$

Subject image $\mathcal{I}$

Result of Tong *et al.*

Our result

Figure 5. Comparison of our result with that of Tong *et al.* [15]. Note that Tong *et al.* employed an additional “before”-makeup example (not shown here), unlike our method.

from others, users may also provide their own previously well-taken photograph, with or without makeup, as the example for retouching. We show an example in Fig. 6. A photo (Fig. 6(a)) with makeup taken in a professional studio is provided as the example. Two other photos of the same person (Fig. 6(b) and (d)) taken by an amateur user are provided as the subject images. In the results (Fig. 6(c) and (e)), the imperfection of the face skin was concealed and pleasant face skin was introduced successfully.

Makeup by portraiture: An interesting application is transferring makeup from a portraiture to a real face photo. This is similar to makeup transfer between real face photos. However, portraitures usually have non-realistic artifacts in the skin detail layer due to the drawing material, or aging and discoloration. These artifacts are not suitable for a face photo. For example, in Fig. 7, the portraiture has some noise. Transferring it makes result unnatural (Fig. 7(c)). If we restrain it by setting $\delta_{\mathcal{E}} = 0$ and $\delta_{\mathcal{I}} = 1$, the result becomes much better (see Fig. 7(d)).

5. Conclusion and Future Work

In this paper, we have presented an approach of creating makeup upon an image with another example image as the makeup style. The main idea is simple yet powerful. Promising results have demonstrated the effectiveness of our approach. One major advantage of our approach is that only one example image is required.

Limitations: The active shape model we adopt assumes

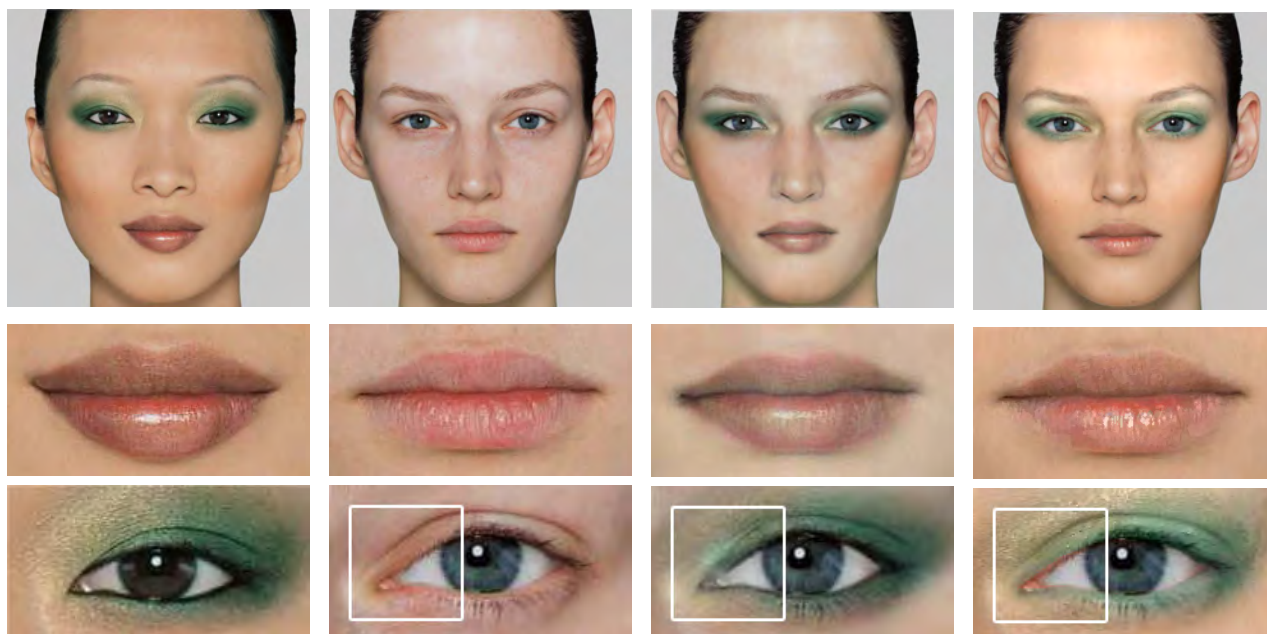
that the face is frontal and upright. Our system is currently tested only with both the subject and example images being nearly frontal. But we envision that it can work well on any pose as long as the pose difference is not large between the subject and example images. One future work is to extend the approach to any pose.

Currently, only one image is used as example. It would be interesting and practical to extend our current approach to multiple example images. In this case, the consistency between different makeup styles may be an issue.

Since only skin detail and color are required from the example image, if we warped these information to a canonical face, the original face structure of the example image is not required any more. Doing so indeed helps to preserve the privacy of the makeup actors. Another future work is to collect more makeup examples and build up a makeup engine enabling users to browse different makeup styles with their own faces, as the previously introduced beauty salon case.

6. Acknowledgments

We thank the reviewers for their valuable comments, Michael S. Brown, Xiaopeng Zhang, Ning Ye, Shaojie Zhuo, and Hao Li for their insightful discussions, as well as Wenzhe Su and Jing Sun for providing their photographs. We acknowledge the generous support of NUS.



Example image $\mathcal{E}$

Subject image $\mathcal{I}$

Result of Tong *et al.*

Our result

图5. 我们的结果与Tong *et al* [15] 的结果对比。需要注意的是, Tong *et al* 使用了一个额外的“前置”美化示例 (此处未展示), 而我们的方法则没有这样做。

从他人处, 用户还可以提供自己之前拍摄的高质量照片 (无论是否化妆) 作为修复示例。图6展示了其中一个例子。一张在专业摄影棚内拍摄并带有妆容的照片 (图6(a)) 被用作参考示例。另外两张由业余用户拍摄的同一个人的照片 (图6(b)和(d)) 则作为待处理的目标图像提供。结果显示 (图6(c)和(e)), 面部皮肤上的瑕疵得到了有效遮盖, 同时引入了令人愉悦的皮肤效果。

通过肖像进行化妆转移: 一个有趣的应用是从一幅肖像转移到一张真实人脸的照片上的化妆。这类类似于在两张真实人脸照片之间进行的化妆转移。然而, 由于绘画材料或老化变色的原因, 肖像通常会在皮肤细节层中包含非真实的瑕疵。这些瑕疵不适合应用到人脸照片上。例如, 在图7中, 肖像有一些噪点。直接将其转移到人脸会导致结果显得不自然 (见图7(c))。如果我们通过设置 $\delta_{\mathcal{E}} = 0$ 和 $\delta_{\mathcal{I}} = 1$ 来加以限制, 则结果会好得多 (参见图7(d))。

5. 结论与未来工作

在本文中, 我们提出了一种基于另一张示例图像作为妆容风格, 在图像上创建妆容的方法。该方法的核心思想简单而强大。令人满意的结果证明了我们方法的有效性。我们的方法的一个主要优势是只需要一张示例图像。

限制: 我们采用的活动形状模型假设

确保人脸是正面且垂直的。我们的系统目前仅在主体和示例图像都接近正面时进行测试。但我们认为, 只要主体与示例图像之间的姿态差异不大, 该系统也能适用于任何姿态。未来的工作之一是将这种方法扩展到处理任意姿态的情况。

目前仅使用了一张图片作为示例。将当前方法扩展到多张示例图片上会很有意义且实用。在这种情况下, 不同妆容风格之间的一致性可能会成为一个问题。

由于只需要从示例图像中提取皮肤细节和颜色信息, 如果我们将这些信息映射到一个标准人脸模型上, 那么原始示例图像的人脸结构就不再需要了。这样做确实有助于保护化妆演员的隐私。另一个未来的工作是收集更多的化妆示例, 并建立一个化妆引擎, 使用户能够用自己的面部浏览不同的妆容风格, 就像之前介绍的美容院案例一样。

6. 致谢

我们感谢审稿人的宝贵意见, 感谢 Michael S. Brown、Xiaopeng Zhang、Ning Ye、Shaojie Zhuo 和 Hao Li 的深入讨论, 同时也感谢 Wenzhe Su 和 Jing Sun 提供照片。我们感谢新加坡国立大学的慷慨支持。

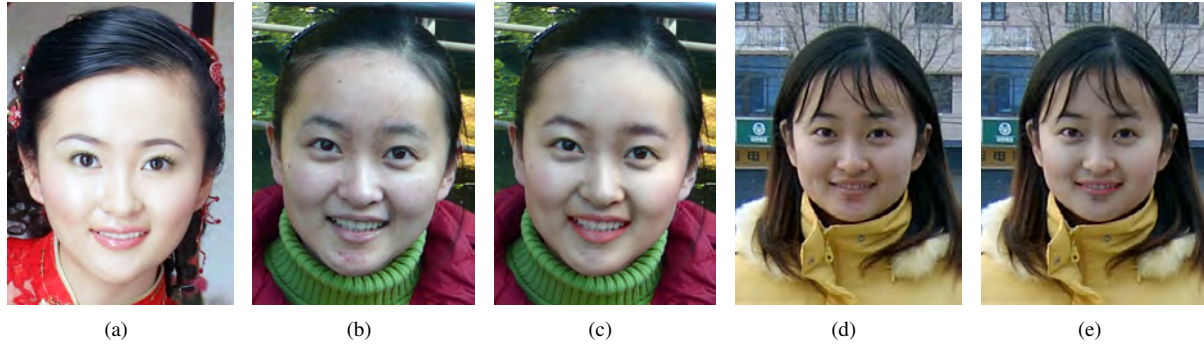


Figure 6. Photo retouching. (a) The example image, taken in a professional studio, provides the desired retouching result. (b)(d) Photos of the same person in (a), taken by an amateur. (c)(e) The retouching results of (b) and (d), respectively.

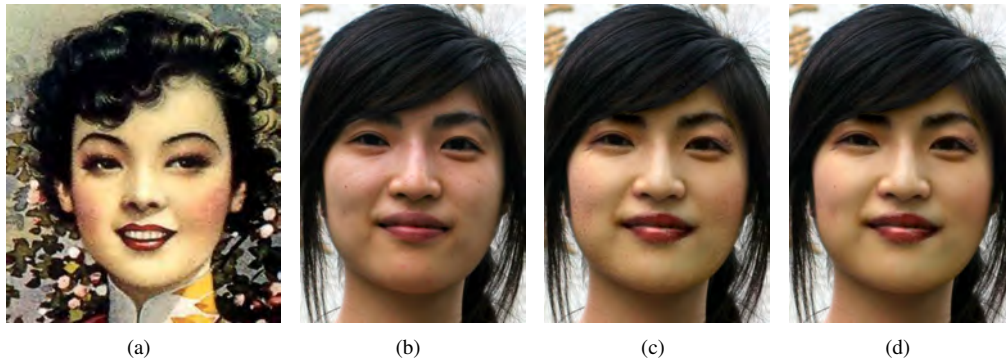


Figure 7. Makeup by portraiture. (a) An old portraiture scan, as the example image. (b) A photo, as the subject image. (c) The makeup result with skin detail from (a). (d) The makeup result with skin detail from (b).

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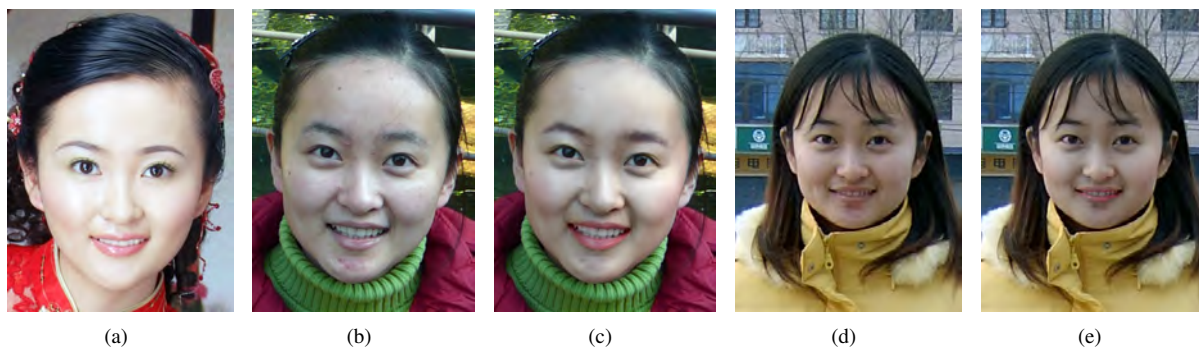


图6. 照片修整。(a) 在专业摄影棚拍摄的示例照片, 展示了期望的修整效果。(b)、(d) 分别为(a)中同一个人由业余摄影师拍摄的照片。(c)、(e) 分别为对(b)和(d)进行修整后的结果。

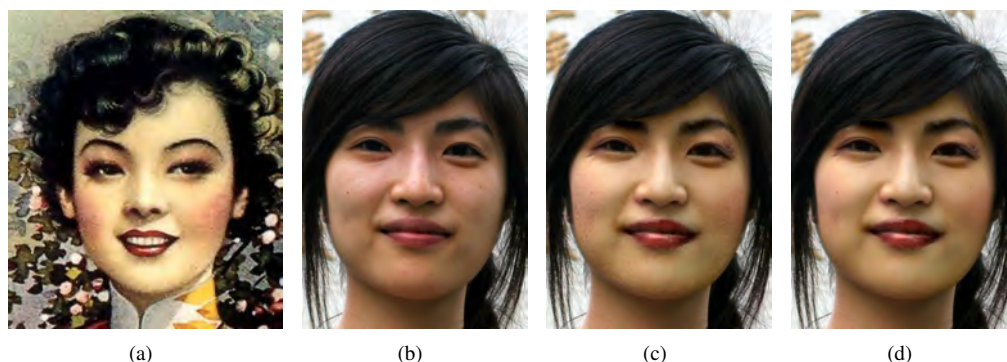


图7. 肖像化妆效果。(a) 作为示例图像的老肖像扫描。(b) 作为主体图像的照片。(c) 从(a)中获得的皮肤细节的化妆结果。(d) 从(b)中获得的皮肤细节的化妆结果。

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